

Hunner-lesion interstitial cystitis after the audit: a preregistered net-of-inflammation residual program with transparent negative results

MEGA-PROGRAM-27, interstitial cystitis lane (builder branch builder-25-ic)

26 September 2026 - working disease manuscript; not yet complete

Abstract

Interstitial cystitis/bladder pain syndrome with Hunner lesions (HIC) is a distinct inflammatory bladder disease whose identification still rests on cystoscopy, not on a molecular marker. This report does four things, in order, and keeps every negative on the record. First, a duplicate-sample and mislabel audit shows the historical top-50 IC expression panel does not survive correction: in the post-audit two-source replication the old and new top-50 lists share zero genes, no gene reaches BH $q < .05$, and the registered sign-agreement threshold fails (43/50, $p = .0469$ against a .025 bar). Second, we acquire and individually verify 98 new accession records, bringing the lane to 153 accession-backed records with honest nesting labels; the statistical evidence behind them rests on 25 paired HIC patients, 13 BCG samples, 13 urine-sediment samples and 20 EV-miRNA subjects, and the record count is an audit figure, not a replication unit. Third, we preregister a module-level test on the 2025 paired Hunner-lesion RNA-seq cohort (GSE238208; 25 patients, lesion and non-lesion biopsies, plus 13 BCG inflammatory-cystitis samples): the lesion program replicates across patients (19/25, $p = .0073$) but is shared with BCG inflammation, so the Hunner-specific gate fails. Fourth, a locked amendment tests the net-of-BCG residual: a BCG-adjusted, G2/M-associated transcriptional residual is reproducible across held-out patients within this cohort (18/25, $p = .0216$). The locked specificity bound is very wide (compatible with no specificity and with strong specificity), and two pre-specified exploratory urinary transfers are inconclusive - one small directionally consistent signal carried substantially by a single high sample, and one non-significant EV-miRNA test. Forty-three external scientific services annotate the program (41 under the most conservative recount). No clinical test, named discovery, or benchmark break is claimed; this is a working research manuscript, not a diagnostic claim.

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1 Scope, verdict and reproducibility

This document is the disease-level manuscript for the interstitial-cystitis lane of MEGA-PROGRAM-27. It is written while the project gates are only partly met, and says so. Current gate state: the 120 accession-record floor is met (153 records, Table 1); the 40 genuinely-used external service floor is met (43 counted, 41 most-conservative, Appendix E); the substantive-page floor for this disease paper is not yet met and is stated honestly where relevant; a positive preregistered replication exists for the residual program (Gate A2.1) but no published-comparator benchmark break and no named new discovery is claimed. Every analysis threshold in this manuscript was committed to the repository before the corresponding test was run; preregistrations are `prereg/IC_P48_preregistration.md` and `prereg/IC_P48_A2_amendment.md`, and every test cites the JSON artifact its numbers come from.

The lane began this phase with 55 IC-tagged records in the shared global manifest (1,508 records across diseases); the P45–P47 acquisition added 98 distinct accession records with no collision against any existing entry, moving the lane to 153 and the global manifest to 1,606. Record units are accession records with verified roles, not independent studies, cohorts or people.

2 Disease background and why the old panel had to be re-audited

Interstitial cystitis/bladder pain syndrome (IC/BPS) splits clinically into Hunner-lesion disease (HIC), an inflammatory phenotype with ulcerative lesions visible at cystoscopy, and non-lesion bladder pain syndrome (BPS), which shares pain but not the lesion biology. BCG intravesical therapy for bladder cancer produces an iatrogenic inflammatory cystitis that is the most important molecular comparator: any “Hunner signature” that is equally present in BCG cystitis is a general severe-inflammation signature, not a Hunner marker. The historical in-program IC panel descended from two array series (GSE621, GSE11783). The program-wide audit found two IC-derived mock-APF samples in GSE621 had been treated as normal controls, and that GSE11783’s ten IC libraries are nested in five patients, so library-level inference was not patient-level. The post-audit sensitivity (Section 4) is therefore the correct statement of the historical panel: it is negative.

3 Data acquisition and provenance

Each GSM record used in this lane was fetched individually (SOFT text retained in `sources/`), mapped to its matrix column by deposited title and description, and labeled with its role, including honest nesting (for example, “not independent cohort/participant” where libraries repeat a person). The P45 acquisition (GSE238208) contributes 64 sample records: 25 paired Hunner lesion and non-lesion biopsies and 13 BCG samples across 38 people. P46 (GSE196156) contributes 21 urinary extracellular-vesicle miRNA records (8 cystitis, 2 BPS, 10 controls, one platform). P47 (GSE11839) contributes 13 cultured-urothelium records (3 IC, 3 controls, two media). Together with the re-audited historical records and the individually checked GSE57560 GSMs, the lane holds 153 accession-backed records (145 sample records in Table 1 plus eight series-level records). Duplicate accessions against the global manifest were checked before appending; the manifest’s unit is a unique accession-backed record, not an independent cohort. The ledger arithmetic is: 55 records already credited to this lane (across GSE621, GSE11783, GSE28242 and GSE57560) plus 98 new record units (64 + 21 + 13 from P45, P46 and P47) equals 153; of these, 145 are sample-level records and eight are series-level records. The 120-record floor is therefore met with a margin of 33, and every record carries the people-count caveats of its source (4 and the provenance subsections) so that record units are never read as independent studies.

3.1 Acquisition audit trail

The acquisition audit (`results/ic_p45_p47_acquisition_audit.json`) records, per series, the exact verification performed before any record entered the manifest. For GSE238208: series SOFT fetched; each of 63 GSM sample sections split and stored individually; deposited `Sample_description` PR identifiers matched 63/63 to named author-FPKM columns; all matched columns fully finite across 27,339 gene rows. For GSE196156: all 20 GSMs present as series-matrix columns, each carrying 464 finite miRNA measurements of 2,632 deposited rows, with the remaining rows empty across all samples - a detection-gated miRNA output, uniform across columns, which is why the measured set is defined as the 464 fully observed rows rather than imputed. For GSE11839: all 12 GSMs present as matrix columns, fully finite across 54,675 rows. New records were checked against the live global manifest for collisions before appending, and the four already-credited series (GSE621, GSE11783, GSE28242, GSE57560) were explicitly not re-counted.

3.2 Per-dataset provenance and verification

GSE238208 (P45; RNA-seq, GPL24676 NovaSeq 6000; iScience 2023 [1]). Twenty-five Hunner-type IC patients contribute paired lesion and non-lesion bladder biopsies (50 GSM), and thirteen BCG-cystitis biopsies form the inflammatory comparator block; 38 people, 63 GSM total. Each of the 63 sample SOFT sections was split and stored individually; the deposited `Sample_description` PR identifiers were matched 63/63 to the named columns of the author FPKM matrix, and every matched column is fully finite across all 27,339 gene rows. This is the primary inference dataset; the unit of inference is the patient, never the library.

GSE196156 (P46; urinary extracellular-vesicle miRNA-seq, GPL29077). Eight cystitis, two BPS and ten control urine-EV profiles (20 GSM). The series matrix holds 2,632 miRNA rows, of which 464 are fully measured across all 20 samples; rows with missing calls are excluded by the locked definition of the measured set, and the group order was verified against the individually stored sample records. The two BPS profiles are retained for description only.

GSE11839 (P47; cultured urothelium array; J. Urology 2008 [2]). Three IC and three control urothelial cultures under two media (13 GSM). Used for acquisition completeness and context; cultured cells are not a lesion surrogate, and no gate is run on this series.

GSE28242 (urine sediment array; J. Urology 2012 [3]). Five control, five without-lesion and three with-lesion urine-sediment profiles (13 GSM), individually re-checked against their sample records. Hosts the exploratory A2.3a transfer; the with-lesion arm is three donors.

GSE57560 (bladder biopsies; J. Urology 2014 [4]). All seven historically analyzed GSMs were individually checked against source title, disease state, bladder-capacity phenotype, platform and matrix column (four low-capacity IC, three controls); they are the previously seen sensitivity series for the re-audit and are never described as fresh validation.

GSE621 [5] and **GSE11783** [6] (historical discovery arrays). The audit found two IC-derived mock-APF samples in GSE621 had been mislabeled as normal controls, and GSE11783's ten IC libraries nest in five patients; both corrections are applied before any re-analysis, and the consequences are reported as negatives.

3.3 Record roles and nesting

Manifest roles are explicit about dependence. GSE238208's 63 GSM are paired or comparator libraries from 38 people and are used as 25 patient pairs plus a 13-sample block, never as 63 independent observations. GSE11783's ten IC libraries nest in five patients and enter the re-audit as such. GSE621 contributes twelve patient samples and two IC-derived mock-APF samples whose historical "normal control" labels were wrong; those two records are kept with their corrected labels and the misclassification is reported as an audit finding. GSE57560's seven records are the previously seen sensitivity set. No record is counted twice, and no library is promoted to a person.

4 The corrected two-source replication is negative

After the audit corrections, discovery was restricted to GSE621 and GSE11783 with mislabeled and nested samples handled, and sensitivity was recomputed in the previously seen GSE57560 (so it cannot be called fresh independent replication). Of 3,326 genes measured in both sources, the new top-50 shares zero genes with the old top-50; the best new-cohort BH value is $q = .418$ with no gene under .05; and the validation sign agreement is 43/50 (86%) against an empirical null mean of 75.1%, $p = .0469$, which fails the registered .025 threshold. The historical positive panel is therefore recorded as a ledger negative, not rewritten into a weaker claim.

4.1 Audit and sensitivity arithmetic, exactly as computed

The reproducibility detail a reviewer needs to rerun this verdict is as follows. The frozen historical result (T1) stands on the prior sample-level discovery: 40/40 sign agreements in the original 6-vs-8 and 10-vs-6 contrasts, left intact as a historical artifact. Post-audit, discovery was restricted to the corrected GSE621 and GSE11783 and required evidence from BOTH studies ($k = 2$), leaving 3,326 genes measured in both. On that intersection: the new top-50 by effect shares zero genes with the old top-50; the best new-cohort Benjamini-Hochberg value is $q = .418$ and no gene falls under .05. The GSE57560 sensitivity then scores the new top-50 in 4 case and 3 control libraries: sign agreement is 43/50 (86%) against an empirical null mean of 75.1% ($p = .0469$), which fails the registered .025 bar, and genes with nominal direction-consistent $p < .05$ number 5 against a null mean of 4.02 ($p = .375$). Both sensitivity statistics sit inside their nulls. Two caveats are kept with the numbers: GSE57560 was previously seen, so this is a sensitivity, never a fresh replication; and a 4-vs-3 library contrast is too small to support any classifier-utility claim even had it passed. A 2025 publication used overlapping studies, which is why the audit trail, not the literature, is the authority for what the old panel does.

5 Preregistered module framework (P48)

The redirect after the negative re-audit was preregistered before any testing. Six candidate biology modules and one known-biology covariate module were locked with fixed gene sets (Table 3) from public libraries (MSigDB Hallmark via Enrichr; GO Biological Process 2025 for the pain module, GO:0019233, 21 genes). The design uses the paired structure of GSE238208: 25 HIC patients each contribute a lesion and a non-lesion biopsy, and the 13 BCG samples form a fixed comparator block.

For training fold T and gene g , per-gene z -scores are computed over the 48 training HIC columns only:

$$z_{g,s} = \frac{x_{g,s} - \bar{x}_{g,T}}{\hat{\sigma}_{g,T}}. \quad (1)$$

Module m 's per-patient paired effect is the mean member-gene z difference:

$$\Delta_{m,i} = \bar{z}_{m,\text{lesion},i} - \bar{z}_{m,\text{nonlesion},i}, \quad e_m = \frac{1}{|T|} \sum_{i \in T} \Delta_{m,i}. \quad (2)$$

A module enters the Gate-1 score with weight +1 iff its mean paired effect is positive and a one-sided paired t -test over training patients gives $p < .05$; the held-out patient score is

$$D_i = \sum_{m: \text{selected}} \Delta_{m,i}. \quad (3)$$

M0 (interferon- γ inflammatory) is an excluded covariate: if M0 alone passes and M1–M6 fails, the result is declared known-inflammatory rediscovery, a novelty negative by construction.

6 Expression processing and verification protocol

The author FPKM matrix of GSE238208 is used as deposited; no re-quantification was attempted, and the deposited values are treated as the study’s own summary. Within each leave-one-patient-out fold, z-scores are recomputed over the 48 training columns only, so no held-out patient contributes to any normalization constant, module weight, or threshold. Module gene sets are used exactly as committed; coverage is reported per module (Table 3) and never silently filled. For the urinary legs, platform annotation is the bottleneck and is therefore evidenced: A2.3a maps M6 symbols through the full GPL6244 platform table (190/190 recovered), and A2.3b maps the 464 measured MIMAT accessions to mature miRNA names through miRBase mature.fa (2,789 human entries). Every fetch URL is logged in the result JSON at analysis time, per the amendment. ChatGPT consults (rounds 1–2) were used in redirect mode only; verbatim prompts and responses are preserved in `redirect/`, all numerical thresholds were recomputed locally, and one incorrect threshold suggested by the consult ($\geq 21/25$) was corrected to $\geq 18/25$ with the correction recorded in `redirect/round2_meta.json`.

6.1 Inference procedures

All gated statistics were computed with the same three primitives. (1) Leave-one-patient-out scoring: for each of the 25 patients in turn, module gene weights and any reference offsets are refit on the other 24 patients, the held-out patient’s lesion and non-lesion libraries are scored without that patient contributing to any parameter, and D is the lesion-minus-non-lesion difference of the module composite. (2) Sign inference: the primary statistic is the count of positive D values, tested by an exact one-sided binomial against $p_0 = 0.5$; the locked pass bar is $\geq 18/25$ with $p < 0.025$ (the count tail, not an asymptotic approximation). (3) Effect-size context: a bootstrap 95% confidence interval on the median D is reported alongside, and for the urine legs - where the design is a group contrast, not a paired sign test - a one-sided permutation null with 20,000 shuffles and fixed seed 20260926. No other inferential machinery is used anywhere in the locked analysis; every number quoted in the gates derives from these primitives applied to the committed fold and composite files.

6.2 Module coverage and symbol nomenclature

Of the 200 committed M6 genes, 190 are present in the author FPKM matrix. The ten absent - H2AX, H2AZ1, H2AZ2, H2BC12, JPT1, KMT5A, KNL1, MAP3K20, NSD2, TENT4A - are older or alias symbol forms (for example H2AX vs H2AFX, JPT1 vs HN1, KNL1 vs CASC5) whose nomenclature does not match the matrix’s symbols. No remapping was attempted after the lock: the committed gene list governs, coverage is reported, and the missing ten are treated as unmeasured rather than recovered by post-lock alias judgment. The same rule produced the other modules’ coverage in Table 3.

7 Gate 1: the lesion axis is reproducible across held-out patients

Leave-one-patient-out over the 25 HIC patients, refitting every quantity per fold, gives 19/25 held-out patients with $D_i > 0$ (exact one-sided binomial $p = .0073$, threshold .025), median $D = 0.273$ with bootstrap 95% CI [0.068, 0.644]. The frozen weight rule selected only module M2 (ECM/EMT remodeling) in the training folds. The M0-only novelty control fails (14/25, $p = .345$), so the pass is not known-inflammatory rediscovery. The frozen rule selected M2 in all 25 of 25 training folds (no other module was ever selected), and held-out D ranges from -0.687 to 1.546 with median 0.273 . Per-fold values are in Appendix B.

8 Gate 2: the reproducible axis is shared with BCG inflammation

Applying the frozen Gate-1 score to the BCG block is negative for Hunner specificity: the score is higher in BCG (0.305) than in Hunner lesions (0.181), Hedges $g = -0.260$, permutation $p = .776$ against the locked pass bar ($g \geq 0.8$, $p < .025$). The replicating lesion axis is therefore severe inflammatory-remodeling biology, not Hunner-specific biology. This is a kept negative, and it motivates the residual design rather than a re-thresholded claim.

9 Amendment A2: net-of-BCG residual program

The amendment was locked before any A2 testing. For module m , the BCG component over the fixed comparator block is

$$b_m = \frac{1}{13} \sum_{s \in \text{BCG}} \bar{z}_{m,s}, \quad (4)$$

and the residual module effect is $A_m = e_m - b_m$. A module enters the residual score with weight +1 iff $A_m > 0$ and the one-sided paired t -test of $\Delta_{m,i} - b_m$ over training patients gives $p < .05$; the held-out residual is

$$D_i^{\text{res}} = \sum_{m: \text{selected}} (\Delta_{m,i} - b_m). \quad (5)$$

BCG never enters a HIC training fold; the block is exchangeable.

10 Gate A2.1: the G2/M residual is reproducible across held-out patients

10.1 Per-patient effect narrative

Across the 25 leave-one-patient-out folds, the Gate 1 lesion-versus-non-lesion contrast (M2) has median $D = 0.273$ and range $[-0.687, +1.546]$, with 19 of 25 patients positive. The strongest individual effects are patients 25 ($D = 1.546$), 18 (1.139) and 13 (1.126); the six negative patients are led in magnitude by 24 (-0.687), 9 (-0.543) and 21 (-0.466). The A2.1 net-of-BCG residual (M6) has median $D_{\text{resid}} = 0.180$ and range $[-0.680, +2.047]$, with 18 of 25 positive. Its three strongest patients are 21 (2.047), 24 (1.158) and 9 (0.888) - precisely the three most negative patients of the M2 contrast. The anti-correspondence is not an artifact of the residual construction: the residual subtracts the fixed BCG offset, identical for every patient, so it cannot invert a per-patient ranking by itself. What the folds show is that the patients whose lesion tissue looks least like inflammatory remodeling are the ones whose tissue shows the strongest G2/M cell-state program relative to BCG inflammation. This is descriptive only; no test was run on the inversion itself.

Only module M6 (G2/M checkpoint, 190 matrix-covered genes) is selected in every fold. The residual contrast meets the locked bar exactly: 18/25 held-out patients positive, $p = .0216$ ($P(X \geq 18 \mid 25, .5) = .02164$), median residual 0.180, bootstrap 95% CI $[0.0027, 0.423]$ excluding zero. M6 was the only module selected in any fold (25/25), and held-out residual values range from -0.680 to 2.047 with median 0.180. Reading: after removing the module-level biology that BCG inflammatory cystitis also induces, Hunner lesions retain a within-patient, BCG-adjusted, G2/M-associated transcriptional residual that is reproducible across held-out patients in this cohort. This is within-cohort patient-level reproducibility, not independent replication.

11 Gate A2.2: specificity bound is compatible, not a validation

The BCG block enters the A2.1 subtraction, so re-testing it cannot be independent; the amendment locks this as a descriptive bound. The residual-adjusted score gives Hedges $g = 0.454$ (95% CI $[-0.144, 1.073]$),

permutation $p = .089$: positive direction, CI overlapping zero - COMPATIBLE, underpowered at 13 BCG samples, and never reported as pass/fail. Stated plainly: the available BCG comparator did not provide evidence sufficient to distinguish the residual from no specificity; the estimate remains compatible with a specificity effect but is not evidence for one, and “bound” here means a very wide interval, $[-0.144, 1.073]$ in Hedges g units.

12 Patient-level complementarity of the two passes (post-hoc, descriptive)

The two passing gates measure different biology, and their failures fall on different people. Gate 1 is negative for patients 1, 9, 10, 14, 21 and 24; Gate A2.1 is negative for patients 4, 6, 11, 13, 16, 17 and 18; the overlap is empty. Every one of the 25 patients is therefore positive in at least one of the two locked scores, 12 are positive in both, and 13 are positive in exactly one. This is a post-hoc description of committed per-fold values (Appendix B), not a preregistered endpoint, and it must not be quoted as a third gate; what it shows is that the remodeling axis and the G2/M-associated residual are not redundant measurements of one program - the patients without a remodeling signal are exactly the patients carrying the residual signal. A complementary observation in the Gate-1 artifact is that the per-patient M2 and M0 paired effects are essentially uncorrelated (Pearson $r = -0.078$, $p = 0.712$), so the remodeling pass is not shadowing the known interferon- γ axis patient-by-patient either.

13 Urinary transfer (exploratory)

Two pre-specified exploratory legs test whether the frozen M6 residual program has a noninvasive correlate; per the amendment, no thresholds were tuned on urine data. In urine sediment mRNA (GSE28242; 190/190 M6 symbols mapped via the GPL6244 platform table), Hunner-PBS urine ($n = 3$) scores above lesion-free/controls ($n = 10$): difference 0.286, one-sided permutation $p = .0406$ - a signal carried substantially by a single high sample (see the per-sample values below), in three donors. In urinary extracellular-vesicle miRNA (GSE196156), the pre-specified validated miRTarBase repressors of the 200-gene M6 module (87,627 human miRNA-target pairs; 262 of the 464 fully measured miRNAs are regulators; the locked direction is lower composite in cases) give composite means -0.0273 (cystitis, $n = 8$) versus $+0.0525$ (control, $n = 10$), difference -0.0798 in the predicted direction, one-sided permutation $p = .1144$. Neither leg meets the .025 bar. A post-hoc descriptive note (`results/ic_p48_a2.urine_descriptive_synthesis.md`) states both directions with exact values and cohort caveats; per the owner ruling it is not a combined test, not a preregistered replication, and no joint p -value is computed.

13.1 Per-sample detail of the two urine legs

In GSE28242 the three with-lesion urines score $+0.480$, $+0.199$ and -0.020 (mean $+0.220$); the ten lesion-free and control urines range from -0.295 to $+0.207$ (mean -0.066), and two of the five control subjects score above $+0.12$. The group difference of 0.286 is therefore carried substantially by one high with-lesion profile (GSM699147) on a three-donor arm; the one-sided permutation $p = .0406$ is the weakest link in the chain and is reported, not rounded up. In GSE196156 the regulator composite is below the control mean ($+0.053$) in six of eight cystitis profiles (range -0.184 to $+0.141$), and four of ten controls also fall below their own group mean - the direction is present but the groups overlap widely, which is exactly what $p = .1144$ says. The two BPS profiles average -0.153 , descriptively at least as far in the predicted case direction as the cystitis arm itself; with $n = 2$ this is anecdote, but it cautions against reading the cystitis-versus-control trend as lesion-specific rather than pain-syndrome-wide.

14 External annotation of the residual program

Forty-three external scientific services were genuinely used on this lane (41 under the most conservative re-count of the NCBI e-utils rows), each with repo-resident evidence (Appendix E). Conclusions drawn from them are contextual, not independent replication, because annotation records share underlying evidence bases. The five M6 genes with the most validated miRNA regulation (G3BP1, SRSF1, DR1, RASAL2, MAPK14) are regulatory hubs rather than a physical complex: STRING returns zero physical edges among them. Their druggability is uneven: DGIdb reports 105 interactions for MAPK14 (including p38 α inhibitors) and none for the other hubs, and Pharos classifies MAPK14 as Tchem while the others are Tbio. In TCGA bladder tumors, mutation records are sparse (RASAL2 16, SRSF1 6, MAPK14 4, G3BP1 2, DR1 1). PubMed abstract co-mention of any hub with “interstitial cystitis” is zero, and Open Targets places none of the hubs among the top-50 targets associated with interstitial cystitis (EFO_1000869): we did not identify prior reports directly testing this locked residual program - a reproducible statement about our searches on this date, not evidence of biological novelty, since annotation coverage is incomplete.

15 Gene-level anatomy of the M6 regulator set

15.1 Platform coverage of the measured regulator slice

The A2.3b regulator universe is large and densely connected: 87,627 validated human miRNA-target pairs name 2,807 distinct regulators against the 200 M6 genes, and per-gene files show a median of 676.5 distinct regulator families per M6 gene. The families targeting at least 100 of the 200 genes are miR-548 (180 genes), miR-29 (172), miR-15 (166), miR-30 (164), miR-16 (162), miR-103 (153), miR-92 (150), miR-17 (150), miR-24 (147) and miR-181 (144) - broadly pleiotropic families, not M6-specific ones. Against that universe, the urine array measures 262 mature forms. These map to 221 distinct miRNA families; 182 of those families contribute a single measured member, and the largest measured family (miR-4433) contributes four. By arm, 162 measured forms are 5p and 100 are 3p. The measured slice is therefore thin and family-sparse relative to the regulator space the module presents, and the A2.3b negative is a statement about that platform’s slice - not about urinary miRNA regulation of the program in general. A panel with denser family coverage would be a different, separately preregisterable test.

The 87,627 validated human miRNA-target pairs behind A2.3b distribute very unevenly across the 200 M6 genes (per-gene counts committed as `results/ic_p48_a2_m6_gene_mirna_pairs.csv`; 87,632 row-level records, of which 87,627 carry hsa- prefixed miRNA names and entered the regulator set). Median coverage is 400.5 pairs per gene. The five most-targeted genes - G3BP1 (1,289), SRSF1 (1,274), DR1 (1,250), RASAL2 (1,220), MAPK14 (1,165) - are regulatory and signaling hubs, not the canonical mitotic machinery: the core G2/M kinases carry markedly less miRNA annotation (CDK1 487, BIRC5 529, AURKA 325, TTK 223, AURKB 203, PLK1 119, CCNB2 59, CDC20 35). Two readings follow. First, the composite used in A2.3b is dominated by heavily annotated hub genes, so its direction reflects regulator coverage of hub biology more than of the mitotic core - one more reason the urine legs stay exploratory. Second, the annotation gradient is itself evidence about where miRNA regulation of the G2/M program has been studied: stress-granule and splicing regulators (G3BP1, SRSF1) and the p38 kinase MAPK14 are far better miRNA-annotated than the checkpoint kinases, which matches their broader multi-pathway roles and explains why DGIdb and Pharos flag only MAPK14 as chemically tractable (Tchem) among the hubs.

16 Published reference points

The published expression literature for Hunner-lesion IC is small, which is why this lane treats every comparison as landscape rather than benchmark. The primary reference point is the source study of GSE238208 [1], which established the interferon- γ /Th1-17 polarization of HIC tissue and motivated the M0 covariate design here: our framework asks what survives after exactly that known axis is excluded, and the M0 nov-

elty control confirms the known axis alone does not reproduce the Gate-1 pass. The older array literature [5, 6, 4] is the basis of the historical panel whose corrected re-audit is reported as negative above. The urine-sediment study behind GSE28242 [3] evaluated noninvasive mRNA markers and is the direct precedent for the A2.3a transfer leg; its with-lesion arm is the same three-donor group that bounds our exploratory signal. No published locked-protocol comparator exists for the residual-program question asked here, so no same-task benchmark win is claimed or implied; a formal published comparator test remains an open gate of the project.

17 Discussion

The biological reading across the locked gates is consistent. Hunner lesions carry a strong remodeling/inflammatory axis (M2) that replicates across patients but is equally a BCG-cystitis program; treating it as a Hunner marker would mislabel severe inflammation generally, and the locked Gate-2 negative prevents exactly that error. Removing the BCG-shared component leaves a BCG-adjusted, G2/M-associated transcriptional residual that is reproducible across held-out patients within this cohort (A2.1). Whether it reflects increased proliferation, a shifted epithelial/immune/stromal cell mixture, or another cell state cannot be resolved from bulk RNA; the hyperproliferation reading is a hypothesis whose direct tests are single-cell or spatial transcriptomics and proliferation-marker immunostaining, not more bulk scoring. The two urinary legs are inconclusive rather than confirmatory. The honest summary is: a within-cohort reproducible tissue residual program; the urinary path is unproven.

17.1 What the M0 control does and does not rule out

The novelty control asks whether a known IFN γ -inflammatory module (M0) alone would have passed the Gate 1 bar. It does not: 14 of 25 patients positive ($p = 0.345$), median $D = 0.012$. So the M2 pass (19/25) is not explained by the generic observation that inflamed tissue looks inflamed - the canonical inflammatory program does not itself separate lesion from non-lesion in this cohort. The post-hoc per-patient correlation between the M2 and M0 effects is $r = -0.078$ ($p = 0.71$), meaning the patients who drive the M2 pass are not the patients with the strongest inflammatory-module contrast. What the M0 control cannot rule out is the reverse direction: non-lesion tissue in a Hunner bladder may itself be inflamed, compressing any inflammation-based contrast. The control's failure is therefore evidence about module identity, not about the biology of the non-lesion state.

17.2 What an independent validation cohort requires

The falsifiable contribution of this work is a claim about what a future cohort must show, and the cohort requirements follow directly from how the current gates are built. First, paired Hunner-lesion and non-lesion tissue from the same bladders, with patient as the unit of inference - the LOPO design exists because libraries are not independent of donors. Second, BCG-exposure metadata recorded per patient, since the residual program is defined against a BCG-inflammation offset and a cohort without exposure data cannot replicate the net-of-BCG construction. Third, the module membership, weights, offset and endpoint locked in a preregistration before the cohort is unblinded or, for a public accession, before its matrix is opened - the A2.3b episode in this project is a worked example of why: two individually exploratory directional results do not license a post-hoc combination, and any combined-urine variant requires an untouched cohort with fixed weights and a fixed endpoint locked in advance. Fourth, coverage: at least 190 of the 200 M6 genes measurable, and a regulator panel denser than 262 forms if the urinary transfer is to be retested. A cohort meeting those four requirements turns the residual program from a replicated-in-one-cohort finding into a testable prediction.

17.3 Module-level reading of the two gates

The two positive gates select different modules for consistent reasons. Gate 1 rewards the largest reproducible lesion-versus-non-lesion contrast, which is the ECM/EMT remodeling module M2: fibrotic remodeling is the dominant within-patient axis in HIC tissue, but the BCG block shows the same axis, so it carries no Hunner specificity. The residual design subtracts exactly this shared component at module level; what remains is M6, a G2/M-associated transcriptional program. That the residual is G2/M-associated rather than barrier-specific (M1), metabolic (M4), stress-related (M5) or pain-related (M3) is itself an empirical result of the locked selection rule, not a modeling choice: in 25/25 folds no other module cleared the residual bar. The pain module M3 (21 genes, GO:0019233) never enters either score, consistent with lesion biology being separable from pain biology in this cohort. The interferon- γ module M0 fails its novelty control, which keeps the positive gates from being reinterpreted as rediscovery of the known Th1/17 axis [1].

18 Limitations

18.1 Cell composition is an untested alternative explanation

Every score in this manuscript is bulk-tissue or bulk-fluid RNA. A lesion-versus-non-lesion biopsy contrast is also a spatial sampling contrast: Hunner lesions are histologically abnormal regions, so a module difference can reflect a changed epithelial fraction, immune infiltrate, stromal abundance, or lesion architecture rather than a changed transcriptional program within cells. Both positive gates are exposed to this: the M2 remodeling axis could be more stroma, and the M6 G2/M-associated residual could be more cycling cells of any type rather than a urothelial state. No deconvolution was performed (the cohort ships expression matrices without matched cell counts), so cell composition stands as an untested alternative explanation, and the single-cell/spatial and immunostaining experiments named in the Discussion are the direct adjudicators.

18.2 Cohort, platform and annotation bounds

Each dataset bounds what can be said. GSE238208 is 25 patients from one study and one platform; the residual replication is therefore a within-study patient-level result, and between-study heterogeneity is unestimated. The BCG comparator is 13 samples, which is why the A2.2 specificity bound can only be “compatible” rather than informative. The GSE28242 with-lesion arm is three donors, so A2.3a’s directional trend has a wide null. The GSE196156 cystitis arm is eight profiles, the composite averages 262 measured regulators - a broad set that dilutes any per-miRNA signal - and 2,168 deposited rows were detection-gated out by the locked measured-set definition. The A2.2 bound is not independent by construction. Per-gene external annotations are correlated through shared evidence bases (UniProt, GO, GEO), so they cannot be summed as independent support. The historical sensitivity reuses the previously seen GSE57560, so it cannot be described as fresh replication. The cultured-cell series GSE11839 is context only. Nothing here is a clinical test, the exploratory urine legs must not be quoted without their threshold failures, the descriptive urine synthesis must not be quoted as a combined test, and the two BPS urine-EV profiles trend in the same direction as the cystitis arm, which any future lesion-specificity claim would have to confront.

Table 1: Individually verified sample records by series and group.

Accession	Series	Title	Group
GSM298203	GSE11783	patient:10–location:healthy–R:01	
GSM298204	GSE11783	patient:11–location:healthy–R:01	
GSM298205	GSE11783	patient:12–location:ni–R:01	
GSM298206	GSE11783	patient:12–location:ulcus–R:01	
GSM298207	GSE11783	patient:13–location:ni–R:01	
GSM298208	GSE11783	patient:13–location:ulcus–R:01	
GSM298209	GSE11783	patient:14–location:ulcus–R:01	
GSM298210	GSE11783	patient:15–location:ni–R:01	
GSM298211	GSE11783	patient:15–location:ulcus–R:01	
GSM298212	GSE11783	patient:2–location:healthy–R:01	
GSM298213	GSE11783	patient:4–location:ni–R:01	
GSM298214	GSE11783	patient:4–location:ulcus–R:01	
GSM298215	GSE11783	patient:5–location:healthy–R:01	
GSM298216	GSE11783	patient:7–location:healthy–R:01	
GSM298217	GSE11783	patient:9–location:healthy–R:01	
GSM402541	GSE11783	patient:14–location:ni–R:01	
GSM299095	GSE11839	human cultured bladder HB DM, subject 8	control cultured urothel
GSM299096	GSE11839	human cultured bladder HB DM, subject 12	control cultured urothel
GSM299097	GSE11839	human cultured bladder HB DM, subject 13	control cultured urothel
GSM299098	GSE11839	human cultured bladder HB KM, subject 8	control cultured urothel
GSM299099	GSE11839	human cultured bladder HB KM, subject 12	control cultured urothel
GSM299100	GSE11839	human cultured bladder HB KM, subject 13	control cultured urothel
GSM299101	GSE11839	human cultured bladder IC DM, subject 1	IC cultured urothelial
GSM299102	GSE11839	human cultured bladder IC DM, subject 3	IC cultured urothelial
GSM299103	GSE11839	human cultured bladder IC DM, subject 10	IC cultured urothelial
GSM299104	GSE11839	human cultured bladder IC KM, subject 1	IC cultured urothelial
GSM299105	GSE11839	human cultured bladder IC KM, subject 3	IC cultured urothelial
GSM299106	GSE11839	human cultured bladder IC KM, subject 10	IC cultured urothelial
GSM5861368	GSE196156	Urine_Cystitis_1	Cystitis
GSM5861369	GSE196156	Urine_Cystitis_2	Cystitis
GSM5861370	GSE196156	Urine_Cystitis_3	Cystitis
GSM5861371	GSE196156	Urine_Cystitis_4	Cystitis
GSM5861372	GSE196156	Urine_Cystitis_5	Cystitis
GSM5861373	GSE196156	Urine_Cystitis_6	Cystitis
GSM5861374	GSE196156	Urine_Cystitis_7	Cystitis
GSM5861375	GSE196156	Urine_Cystitis_8	Cystitis
GSM5861376	GSE196156	Urine_BPS_1	BPS
GSM5861377	GSE196156	Urine_BPS_2	BPS
GSM5861378	GSE196156	Urine_Control_1	Control
GSM5861379	GSE196156	Urine_Control_2	Control
GSM5861380	GSE196156	Urine_Control_3	Control
GSM5861381	GSE196156	Urine_Control_4	Control
GSM5861382	GSE196156	Urine_Control_5	Control
GSM5861383	GSE196156	Urine_Control_6	Control
GSM5861384	GSE196156	Urine_Control_7	Control
GSM5861385	GSE196156	Urine_Control_8	Control
GSM5861386	GSE196156	Urine_Control_9	Control
GSM5861387	GSE196156	Urine_Control_10	Control
GSM7660353	GSE238208	Hunner_Type_Interstitial_Cystitis_Case1_Hunner_Lesion	Hunner lesion
GSM7660354	GSE238208	Hunner_Type_Interstitial_Cystitis_Case2_Hunner_Lesion	Hunner lesion
GSM7660355	GSE238208	Hunner_Type_Interstitial_Cystitis_Case3_Hunner_Lesion	Hunner lesion
GSM7660356	GSE238208	Hunner_Type_Interstitial_Cystitis_Case4_Hunner_Lesion	Hunner lesion
GSM7660357	GSE238208	Hunner_Type_Interstitial_Cystitis_Case5_Hunner_Lesion	Hunner lesion
GSM7660358	GSE238208	Hunner_Type_Interstitial_Cystitis_Case6_Hunner_Lesion	Hunner lesion
GSM7660359	GSE238208	Hunner_Type_Interstitial_Cystitis_Case7_Hunner_Lesion	Hunner lesion
GSM7660360	GSE238208	Hunner_Type_Interstitial_Cystitis_Case8_Hunner_Lesion	Hunner lesion
GSM7660361	GSE238208	Hunner_Type_Interstitial_Cystitis_Case9_Hunner_Lesion	Hunner lesion
GSM7660362	GSE238208	Hunner_Type_Interstitial_Cystitis_Case10_Hunner_Lesion	Hunner lesion
GSM7660363	GSE238208	Hunner_Type_Interstitial_Cystitis_Case11_Hunner_Lesion	Hunner lesion
GSM7660364	GSE238208	Hunner_Type_Interstitial_Cystitis_Case12_Hunner_Lesion	Hunner lesion
GSM7660365	GSE238208	Hunner_Type_Interstitial_Cystitis_Case13_Hunner_Lesion	Hunner lesion

Table 3: Locked modules (gene counts; matrix coverage in parentheses).

Module	Source term	<i>n</i> genes	Role
M0_known_inflammatory	Interferon Gamma Response	200	known-biology covariate; EXCLUDED from the novelty score (EX
M1_barrier_apical_junction	Apical Junction	200 (193 in matrix)	candidate novelty module: epithelial barrier
M2_ecm_remodeling	Epithelial Mesenchymal Transition	200 (197 in matrix)	candidate novelty module: ECM/remodeling
M3_neuronal_sensory_pair	Sensory Perception of Pain (GO:0019233)	21 (20 in matrix)	candidate novelty module: neuronal/sensory
M4_metabolism_oxphos	Oxidative Phosphorylation	200 (184 in matrix)	candidate novelty module: metabolism
M5_stress_repair_hypoxia	Hypoxia	200 (192 in matrix)	candidate novelty module: stress/repair
M6_cell_state_g2m	G2-M Checkpoint	200 (190 in matrix)	candidate novelty module: cell-state/proliferation

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A Derivations

The binomial gate uses the exact one-sided tail. For n independent patients with null success probability $1/2$,

$$P(X \geq k | n) = \sum_{j=k}^n \binom{n}{j} 2^{-n}, \quad (6)$$

so $P(X \geq 19 | 25) = 0.00732$ and $P(X \geq 18 | 25) = 0.02164$; the locked bar is .025 one-sided, matching the program's two-sided .05 convention. Hedges' g corrects the standardized mean difference for small samples:

$$d = \frac{\bar{x}_1 - \bar{x}_0}{s_p}, \quad g = Jd, \quad J = 1 - \frac{3}{4(n_1 + n_0) - 9}. \quad (7)$$

The permutation p for a one-sided mean difference Δ over B label reshuffles is the $(+1)$ -smoothed tail fraction

$$p = \frac{1 + \#\{b : \Delta_b \text{ at least as extreme as } \Delta_{\text{obs}}\}}{1 + B}, \quad (8)$$

which is never zero and is exact for the exchangeable null. The residual construction (Eq. 5–6) is a module-space analogue of partialling out a comparator-induced component: because b_m is a constant within a fold, the paired t -test of $\Delta_{m,i} - b_m$ is the paired test of $\Delta_{m,i}$ against the shifted null b_m , so BCG contributes a fixed offset, never variance, to the patient-level inference.

A.1 A fully worked fold: patient 25 under Gate 1

One complete fold, end to end, from the committed values (`results/ic_p48_gate1_fold_detail.csv`). Patient 25 is held out; module weights and reference distributions are refit on the other 24 patients with the locked procedure, and the held-out pair is scored with no parameter seeing patient 25. The M2 composite scores are $S_{\text{lesion}} = +0.9777$ and $S_{\text{nonlesion}} = -0.5683$, giving $D = +1.5460$ - the largest positive fold in the gate (range $[-0.6874, +1.5460]$). The fold contributes one to the count of positives; the count reaches 19, and the exact one-sided binomial tail $P(X \geq 19 | n = 25, p = 0.5) = 0.00732$ is the Gate 1 p -value. Under A2.1 the same held-out machinery on the BCG-adjusted residual gives patient 21 the largest value, $D_{\text{resid}} = +2.0467$ (median 0.1798), and the count of positives is 18 with tail $P(X \geq 18 | 25, .5) = 0.02164$. No other statistic enters either gate.

A.2 Worked tail computations

With $n = 25$ paired patients and null per-patient success probability $1/2$, the exact one-sided Gate-1 tail is

$$P(X \geq 19 | 25) = 2^{-25} \sum_{j=19}^{25} \binom{25}{j} = \frac{184,756 + 53,130 + 23,000 + 5,200 + 2,300 + 126 + 25 + 1 \text{ terms}}{33,554,432} = 0.00732. \quad (9)$$

The A2.1 tail is $P(X \geq 18 | 25) = 0.02164$; both are below the locked .025 bar, while $P(X \geq 17 | 25) = 0.0539$ would not have been - so the locked threshold $k = 18$ is the smallest passing count, and the M0-only control ($14/25, p = 0.345$) fails by a wide margin. These integers, not asymptotics, are the test.

A.3 Why the permutation null is the right null here

The LOPO sign test assumes only that, under no lesion program, the sign of each held-out D_i is an independent fair coin; pairing removes all between-patient expression level differences by construction. For the two-sample permutation tests (Gate 2, A2.2, both urine legs), the null is exchangeability of group labels; the BCG block is a fixed comparator in the amendment design, so in A2.2 only the spread of the bound is informative, and the amendment locks that reading before computation.

A.4 The residual as a fixed offset

Because b_m is constant within a fold, testing $\Delta_{m,i} - b_m > 0$ over patients is the location-shifted paired test of $\Delta_{m,i}$ against b_m :

$$\text{Var}_i(\Delta_{m,i} - b_m) = \text{Var}_i(\Delta_{m,i}), \quad t = \frac{\overline{\Delta_m} - b_m}{s_{\Delta_m} / \sqrt{n_T}}, \quad (10)$$

so the BCG block contributes a fixed offset to the numerator and nothing to the variance. This is the module-space analogue of partialling out a comparator-induced component, and it is why A2.2 cannot recycle the same block as an independent validation set.

B Per-fold held-out values

Gate 1 folds

held_out_patient	modules_used	S_lesion	S_nonlesion	D
1	M2_ecm_modeling	-0.4105	-0.2845	-0.126
2	M2_ecm_modeling	0.895	-0.1606	1.0555
3	M2_ecm_modeling	-0.3774	-0.8103	0.4329
4	M2_ecm_modeling	0.3802	-0.6533	1.0335
5	M2_ecm_modeling	0.4063	0.187	0.2193
6	M2_ecm_modeling	0.4532	-0.1544	0.6076
7	M2_ecm_modeling	0.6771	0.4041	0.273
8	M2_ecm_modeling	-0.0886	-0.1571	0.0685
9	M2_ecm_modeling	-0.5234	0.02	-0.5434
10	M2_ecm_modeling	0.2467	0.6331	-0.3864
11	M2_ecm_modeling	-0.0354	-0.7767	0.7414
12	M2_ecm_modeling	0.3245	0.2752	0.0494
13	M2_ecm_modeling	0.4563	-0.6693	1.1256
14	M2_ecm_modeling	-0.8056	-0.7532	-0.0524
15	M2_ecm_modeling	-0.1797	-0.3765	0.1968
16	M2_ecm_modeling	0.6742	0.0303	0.6439
17	M2_ecm_modeling	-0.7431	-0.9393	0.1961
18	M2_ecm_modeling	0.8501	-0.2891	1.1392
19	M2_ecm_modeling	0.8193	0.2041	0.6152
20	M2_ecm_modeling	1.0216	-0.0016	1.0232
21	M2_ecm_modeling	-0.1576	0.3086	-0.4662
22	M2_ecm_modeling	0.3673	-0.089	0.4563
23	M2_ecm_modeling	0.1446	0.0391	0.1055
24	M2_ecm_modeling	-0.6606	0.0268	-0.6874
25	M2_ecm_modeling	0.9777	-0.5683	1.546

Gate A2.1 folds

held_out_patient	modules_used	D_residual
1	M6_cell_state_g2m	0.0027
2	M6_cell_state_g2m	0.5956
3	M6_cell_state_g2m	0.7106
4	M6_cell_state_g2m	-0.6798
5	M6_cell_state_g2m	0.1401

held_out_patient	modules_used	D_residual
6	M6_cell_state_g2m	-0.0815
7	M6_cell_state_g2m	0.129
8	M6_cell_state_g2m	0.1798
9	M6_cell_state_g2m	0.8881
10	M6_cell_state_g2m	0.501
11	M6_cell_state_g2m	-0.5039
12	M6_cell_state_g2m	0.471
13	M6_cell_state_g2m	-0.0189
14	M6_cell_state_g2m	0.4234
15	M6_cell_state_g2m	0.1956
16	M6_cell_state_g2m	-0.0029
17	M6_cell_state_g2m	-0.2276
18	M6_cell_state_g2m	-0.6137
19	M6_cell_state_g2m	0.2694
20	M6_cell_state_g2m	0.1531
21	M6_cell_state_g2m	2.0467
22	M6_cell_state_g2m	0.2421
23	M6_cell_state_g2m	0.146
24	M6_cell_state_g2m	1.1582
25	M6_cell_state_g2m	0.29

C Locked module gene sets

Gene lists as committed in `prereg/ic_p48_module_genesets.json`.

M0.known_inflammatory_IFNg (200 genes, Interferon Gamma Response)

ADAR	APOL6	ARID5B	ARL4A
AUTS2	B2M	BANK1	BATF2
BPGM	BST2	BTG1	C1R
C1S	CASP1	CASP3	CASP4
CASP7	CASP8	CCL2	CCL5
CCL7	CD274	CD38	CD40
CD69	CD74	CD86	CDKN1A
CFB	CFH	CIITA	CMKLR1
CMPK2	CMTR1	CSF2RB	CXCL10
CXCL11	CXCL9	DDX58	DDX60
DHX58	EIF2AK2	EIF4E3	EPSTI1
FAS	FCGR1A	FGL2	FPR1
GBP4	GBP6	GCH1	GPR18
GZMA	HELZ2	HERC6	HIF1A
HLA-A	HLA-B	HLA-DMA	HLA-DQA1
HLA-DRB1	HLA-G	ICAM1	IDO1
IFI27	IFI30	IFI35	IFI44
IFI44L	IFIH1	IFIT1	IFIT2
IFIT3	IFITM2	IFITM3	IFNAR2
IL10RA	IL15	IL15RA	IL18BP
IL2RB	IL4R	IL6	IL7
IRF1	IRF2	IRF4	IRF5

IRF7	IRF8	IRF9	ISG15
ISG20	ISOC1	ITGB7	JAK2
KLRK1	LAP3	LATS2	LCP2
LGALS3BP	LY6E	LYSMD2	MARCHF1
METTL7B	MT2A	MTHFD2	MVP
MX1	MX2	MYD88	NAMPT
NCOA3	NFKB1	NFKBIA	NLRC5
NMI	NOD1	NUP93	OAS2
OAS3	OASL	OGFR	P2RY14
PARP12	PARP14	PDE4B	PELI1
PFKP	PIM1	PLA2G4A	PLSCR1
PML	PNP	PNPT1	PSMA2
PSMA3	PSMB10	PSMB2	PSMB8
PSMB9	PSME1	PSME2	PTGS2
PTPN1	PTPN2	PTPN6	RAPGEF6
RBCK1	RIPK1	RIPK2	RNF213
RNF31	RSAD2	RTP4	SAMD9L
SAMHD1	SECTM1	SELP	SERPING1
SLAMF7	SLC25A28	SOCS1	SOCS3
SOD2	SP110	SPPL2A	SRI
SSPN	ST3GAL5	ST8SIA4	STAT1
STAT2	STAT3	STAT4	TAP1
TAPBP	TDRD7	TNFAIP2	TNFAIP3
TNFAIP6	TNFSF10	TOR1B	TRAFD1
TRIM14	TRIM21	TRIM25	TRIM26
TXNIP	UBE2L6	UPP1	USP18
VAMP5	VAMP8	VCAM1	WARS1
XAF1	XCL1	ZBP1	ZNFX1

M1_barrier_apical_junction (200 genes, Apical Junction)

ACTA1	ACTB	ACTC1	ACTG1
ACTG2	ACTN1	ACTN2	ACTN3
ACTN4	ADAM15	ADAM23	ADAM9
ADAMTS5	ADRA1B	AKT2	AKT3
ALOX15B	AMH	AMIGO1	AMIGO2
ARHGEF6	ARPC2	ATP1A3	B4GALT1
BAIAP2	BMP1	CADM2	CADM3
CALB2	CAP1	CD209	CD274
CD276	CD34	CD86	CD99
CDH1	CDH11	CDH15	CDH3
CDH4	CDH6	CDH8	CDK8
CDSN	CERCAM	CLDN11	CLDN14
CLDN15	CLDN18	CLDN19	CLDN4
CLDN5	CLDN6	CLDN7	CLDN8
CLDN9	CNN2	CNTN1	COL16A1
COL17A1	COL9A1	CRAT	CRB3
CTNNA1	CTNND1	CX3CL1	DHX16
DLG1	DMP1	DSC1	DSC3
EGFR	EPB41L2	EVL	EXOC4
FBN1	FLNC	FSCN1	FYB1

GAMT	GNAI1	GNAI2	GRB7
GTF2F1	HADH	HRAS	ICAM1
ICAM2	ICAM4	ICAM5	IKBKG
INPPL1	INSIG1	IRS1	ITGA10
ITGA2	ITGA3	ITGA9	ITGB1
ITGB4	JAM3	JUP	KCNH2
KRT31	LAMA3	LAMB3	LAMC2
LAYN	LDLRAP1	LIMA1	MADCAM1
MAP3K20	MAP4K2	MAPK11	MAPK13
MAPK14	MDK	MMP2	MMP9
MPP5	MPZL1	MPZL2	MSN
MVD	MYH10	MYH9	MYL12B
MYL9	NECTIN1	NECTIN2	NECTIN3
NECTIN4	NEGR1	NEXN	NF1
NF2	NFASC	NLGN2	NLGN3
NRAP	NRTN	NRXN2	PARD6G
PARVA	PBX2	PCDH1	PDZD3
PECAM1	PFN1	PIK3CB	PIK3R3
PKD1	PLCG1	PPP2R2C	PTEN
PTK2	PTPRC	RAC2	RASA1
RHOF	RRAS	RSU1	SDC3
SGCE	SHC1	SHROOM2	SIRPA
SKAP2	SLC30A3	SLIT2	SORBS3
SPEG	SRC	STX4	SYK
SYMPK	TAOK2	TGFBI	THBS3
THY1	TIAL1	TJP1	TMEM8B
TNFRSF11B	TRAF1	TRO	TSC1
TSPAN4	TUBG1	VASP	VAV2
VCAM1	VCAN	VCL	VWF
WASL	WNK4	YWHAH	ZYX

M2_ecm_remodeling (200 genes, Epithelial Mesenchymal Transition)

ABI3BP	ACTA2	ADAM12	ANPEP
APLP1	AREG	BASP1	BDNF
BGN	BMP1	CADM1	CALD1
CALU	CAP2	CAPG	CCN1
CCN2	CD44	CD59	CDH11
CDH2	CDH6	COL11A1	COL12A1
COL16A1	COL1A1	COL1A2	COL3A1
COL4A1	COL4A2	COL5A1	COL5A2
COL5A3	COL6A2	COL6A3	COL7A1
COL8A2	COLGALT1	COMP	COPA
CRLF1	CTHRC1	CXCL1	CXCL12
CXCL6	CXCL8	DAB2	DCN
DKK1	DPYSL3	DST	ECM1
ECM2	EDIL3	EFEMP2	ELN
EMP3	ENO2	FAP	FAS
FBLN1	FBLN2	FBLN5	FBN1
FBN2	FERMT2	FGF2	FLNA
FMOD	FN1	FOXC2	FSTL1

FSTL3	FUCA1	FZD8	GADD45A
GADD45B	GAS1	GEM	GJA1
GLIPR1	GPC1	GPX7	GREM1
HTRA1	ID2	IGFBP2	IGFBP3
IGFBP4	IL15	IL32	IL6
INHBA	ITGA2	ITGA5	ITGAV
ITGB1	ITGB3	ITGB5	JUN
LAMA1	LAMA2	LAMA3	LAMC1
LAMC2	LGALS1	LOX	LOXL1
LOXL2	LRP1	LRRC15	LUM
MAGEE1	MATN2	MATN3	MCM7
MEST	MFAP5	MGP	MMP1
MMP14	MMP2	MMP3	MSX1
MXRA5	MYL9	MYLK	NID2
NNMT	NOTCH2	NT5E	NTM
OXTR	P3H1	PCOLCE	PCOLCE2
PDGFRB	PDLIM4	PFN2	PLAUR
PLOD1	PLOD2	PLOD3	PMEPA1
PMP22	POSTN	PIIB	PRRX1
PRSS2	PTHLH	PTX3	PVR
QSOX1	RGS4	RHOB	SAT1
SCG2	SDC1	SDC4	SERPINE1
SERPINE2	SERPINH1	SFRP1	SFRP4
SGCB	SGCD	SGCG	SLC6A8
SLIT2	SLIT3	SNAI2	SNTB1
SPARC	SPOCK1	SPP1	TAGLN
TFPI2	TGFB1	TGFB1	TGFBR3
TGM2	THBS1	THBS2	THY1
TIMP1	TIMP3	TNC	TNFAIP3
TNFRSF11B	TNFRSF12A	TPM1	TPM2
TPM4	VCAM1	VCAN	VEGFA
VEGFC	VIM	WIPF1	WNT5A

M3_neuronal_sensory_pain (21 genes, Sensory Perception of Pain (GO:0019233))

CCL2	CCR2	CHRNA4	CHRNA2
KCNA2	KCNK4	MMP12	MRGPRX2
NIPSNAP1	OPRK1	OPRL1	OPRM1
P2RX4	P2RX7	PRDM12	PROK2
PTGES	RETREG1	SCN11A	SCN9A
TRPA1			

M4_metabolism_oxphos (200 genes, Oxidative Phosphorylation)

ABCB7	ACAA1	ACAA2	ACADM
ACADSB	ACADVL	ACAT1	ACO2
AFG3L2	AIFM1	ALAS1	ALDH6A1
ATP1B1	ATP5F1A	ATP5F1B	ATP5F1C
ATP5F1D	ATP5F1E	ATP5MC1	ATP5MC2
ATP5MC3	ATP5ME	ATP5MF	ATP5MG

ATP5PB	ATP5PD	ATP5PF	ATP5PO
ATP6AP1	ATP6V0B	ATP6V0C	ATP6V0E1
ATP6V1C1	ATP6V1D	ATP6V1E1	ATP6V1F
ATP6V1G1	ATP6V1H	BAX	BCKDHA
BDH2	CASP7	COX10	COX11
COX15	COX17	COX4I1	COX5A
COX5B	COX6A1	COX6B1	COX6C
COX7A2	COX7A2L	COX7B	COX7C
COX8A	CPT1A	CS	CYB5A
CYB5R3	CYC1	CYCS	DECR1
DLAT	DLD	DLST	ECH1
ECHS1	ECI1	ETFA	ETFB
ETFDH	FDX1	FH	FXN
GLUD1	GOT2	GPI	GPX4
GRPEL1	HADHA	HADHB	HCCS
HSD17B10	HSPA9	HTRA2	IDH1
IDH2	IDH3A	IDH3B	IDH3G
IMMT	ISCA1	ISCU	LDHA
LDHB	LRPPRC	MAOB	MDH1
MDH2	MFN2	MGST3	MPC1
MRPL11	MRPL15	MRPL34	MRPL35
MRPS11	MRPS12	MRPS15	MRPS22
MRPS30	MTRF1	MTRR	MTX2
NDUFA1	NDUFA2	NDUFA3	NDUFA4
NDUFA5	NDUFA6	NDUFA7	NDUFA8
NDUFA9	NDUFAB1	NDUFB1	NDUFB2
NDUFB3	NDUFB4	NDUFB5	NDUFB6
NDUFB7	NDUFB8	NDUFC1	NDUFC2
NDUFS1	NDUFS2	NDUFS3	NDUFS4
NDUFS6	NDUFS7	NDUFS8	NDUFV1
NDUFV2	NNT	NQO2	OAT
OGDH	OPA1	OXA1L	PDHA1
PDHB	PDHX	PDK4	PDP1
PHB2	PHYH	PMPCA	POLR2F
POR	PRDX3	RETSAT	RHOT1
RHOT2	SDHA	SDHB	SDHC
SDHD	SLC25A11	SLC25A12	SLC25A20
SLC25A3	SLC25A4	SLC25A5	SLC25A6
SUCLA2	SUCLG1	SUPV3L1	SURF1
TCIRG1	TIMM10	TIMM13	TIMM17A
TIMM50	TIMM8B	TIMM9	TOMM22
TOMM70	UQCR10	UQCR11	UQCRB
UQCRC1	UQCRC2	UQCRCFS1	UQCRCR
UQCRCQ	VDAC1	VDAC2	VDAC3

M5_stress_repair_hypoxia (200 genes, Hypoxia)

ACKR3	ADM	ADORA2B	AK4
AKAP12	ALDOA	ALDOB	ALDOC
AMPD3	ANGPTL4	ANKZF1	ANXA2
ATF3	ATP7A	B3GALT6	B4GALNT2

BCAN	BCL2	BGN	BHLHE40
BNIP3L	BRS3	BTG1	CA12
CASP6	CAV1	CAVIN1	CAVIN3
CCN1	CCN2	CCN5	CCNG2
CDKN1A	CDKN1B	CDKN1C	CHST2
CHST3	CITED2	COL5A1	CP
CSRP2	CXCR4	DCN	DDIT3
DDIT4	DPYSL4	DTNA	DUSP1
EDN2	EFNA1	EFNA3	EGFR
ENO1	ENO2	ENO3	ERO1A
ERRFI1	ETS1	EXT1	F3
FAM162A	FBP1	FOS	FOSL2
FOXO3	GAA	GALK1	GAPDH
GAPDHS	GBE1	GCK	GCNT2
GLRX	GPC1	GPC3	GPC4
GPI	GRHR	GYS1	HAS1
HDLBP	HEXA	HK1	HK2
HMOX1	HOXB9	HS3ST1	HSPA5
IDS	IER3	IGFBP1	IGFBP3
IL6	ILVBL	INHA	IRS2
ISG20	JMJD6	JUN	KDEL3
KDM3A	KIF5A	KLF6	KLF7
KLHL24	LALBA	LARGE1	LDHA
LDHC	LOX	LXN	MAFF
MAP3K1	MIF	MT1E	MT2A
MXI1	MYH9	NAGK	NCAN
NDRG1	NDST1	NDST2	NEDD4L
NFIL3	NOCT	NR3C1	P4HA1
P4HA2	PAM	PCK1	PDGFB
PDK1	PDK3	PFKFB3	PFKL
PFKP	PGAM2	PGF	PGK1
PGM1	PGM2	PHKG1	PIM1
PKLR	PKP1	PLAC8	PLAUR
PLIN2	PNRC1	PPARGC1A	PPFIA4
PPP1R15A	PPP1R3C	PRDX5	PRKCA
PYGM	RBPJ	RORA	RRAGD
S100A4	SAP30	SCARB1	SDC2
SDC3	SDC4	SELENBP1	SERPINE1
SIAH2	SLC25A1	SLC2A1	SLC2A3
SLC2A5	SLC37A4	SLC6A6	SRPX
STBD1	STC1	STC2	SULT2B1
TES	TGFB3	TGFBI	TGM2
TIPARP	TKTL1	TMEM45A	TNFAIP3
TPBG	TPD52	TPI1	TPST2
UGP2	VEGFA	VHL	VLDLR
WSB1	XPNPEP1	ZFP36	ZNF292

M6_cell_state_g2m (200 genes, G2-M Checkpoint)

ABL1	AMD1	ARID4A	ATF5
ATRX	AURKA	AURKB	BARD1

BCL3	BIRC5	BRCA2	BUB1
BUB3	CASP8AP2	CBX1	CCNA2
CCNB2	CCND1	CCNF	CCNT1
CDC20	CDC25A	CDC25B	CDC27
CDC45	CDC6	CDC7	CDK1
CDK4	CDKN1B	CDKN2C	CDKN3
CENPA	CENPE	CENPF	CHAF1A
CHEK1	CHMP1A	CKS1B	CKS2
CTCF	CUL1	CUL3	CUL4A
CUL5	DBF4	DDX39A	DKC1
DMD	DR1	DTYMK	E2F1
E2F2	E2F3	E2F4	EFNA5
EGF	ESPL1	EWSR1	EXO1
EZH2	FANCC	FBXO5	FOXN3
G3BP1	GINS2	GSPT1	H2AX
H2AZ1	H2AZ2	H2BC12	HIF1A
HIRA	HMGA1	HMGB3	HMGN2
HMMR	HNRNPD	HNRNPU	HOXC10
HSPA8	HUS1	ILF3	INCENP
JPT1	KATNA1	KIF11	KIF15
KIF20B	KIF22	KIF23	KIF2C
KIF4A	KIF5B	KMT5A	KNL1
KPNA2	KPNB1	LBR	LIG3
LMNB1	MAD2L1	MAP3K20	MAPK14
MARCKS	MCM2	MCM3	MCM5
MCM6	MEIS1	MEIS2	MKI67
MNAT1	MT2A	MTF2	MYBL2
MYC	NASP	NCL	NDC80
NEK2	NOLC1	NOTCH2	NSD2
NUMA1	NUP50	NUP98	NUSAP1
ODC1	ODF2	ORC5	ORC6
PAFAH1B1	PBK	PDS5B	PLK1
PLK4	PML	POLA2	POLE
POLQ	PRC1	PRIM2	PRMT5
PRPF4B	PTTG1	PTTG3P	PURA
RACGAP1	RAD21	RAD23B	RAD54L
RASAL2	RBL1	RBM14	RPA2
RPS6KA5	SAP30	SFPQ	SLC12A2
SLC38A1	SLC7A1	SLC7A5	SMAD3
SMARCC1	SMC1A	SMC2	SMC4
SNRPD1	SQLE	SRSF1	SRSF10
SRSF2	SS18	STAG1	STIL
STMN1	SUV39H1	SYNCRIP	TACC3
TENT4A	TFDP1	TGFB1	TLE3
TMPO	TNPO2	TOP1	TOP2A
TPX2	TRA2B	TRAIP	TROAP
TTK	UBE2C	UBE2S	UCK2
UPF1	WRN	XPO1	YTHDC1

D Pre-specified miRNA regulators (A2.3b)

Named human regulators of the M6 module from miRTarBase v10 (87,627 validated pairs); 262 mature miRNAs measured in GSE196156.

hsa-let-7-5p/98-5p	hsa-let-7a-2-3p
hsa-let-7a-5p	hsa-let-7b-3p
hsa-let-7d-3p	hsa-let-7d-5p
hsa-let-7f-2-3p	hsa-let-7f-2-3p/1185-3p
hsa-let-7i-3p	hsa-let-7i-5p
hsa-miR-10-5p	hsa-miR-100-3p
hsa-miR-101-3p.1	hsa-miR-101-3p.2
hsa-miR-103a-1-5p	hsa-miR-103a-2-5p
hsa-miR-105-3p	hsa-miR-105-5p
hsa-miR-106a-5p	hsa-miR-106b-3p
hsa-miR-10a-5p	hsa-miR-10b-3p
hsa-miR-1179	hsa-miR-1180-3p
hsa-miR-1184	hsa-miR-1185-1-3p
hsa-miR-1197	hsa-miR-1199-3p
hsa-miR-1203	hsa-miR-1204
hsa-miR-1207-5p	hsa-miR-1207-5p/4763-3p
hsa-miR-1224-3p	hsa-miR-1224-5p
hsa-miR-1226-5p	hsa-miR-1227-3p
hsa-miR-1229-3p	hsa-miR-1229-5p
hsa-miR-1233-5p/6778-5p	hsa-miR-1234-3p
hsa-miR-1237-3p	hsa-miR-1237-5p
hsa-miR-124-3p	hsa-miR-124-3p.1
hsa-miR-1244	hsa-miR-1245a
hsa-miR-1246	hsa-miR-1247-3p
hsa-miR-1249-5p	hsa-miR-1249-5p/6797-5p
hsa-miR-1251-3p	hsa-miR-1251-5p
hsa-miR-1254/3116	hsa-miR-1255a
hsa-miR-1256	hsa-miR-1257
hsa-miR-125b-1-3p	hsa-miR-125b-2-3p
hsa-miR-126-3p.2	hsa-miR-126-5p
hsa-miR-1261	hsa-miR-1262
hsa-miR-1265	hsa-miR-1266-3p
hsa-miR-1268a/1268b	hsa-miR-1269
hsa-miR-127-5p	hsa-miR-1270
hsa-miR-1273a	hsa-miR-1273c
hsa-miR-1273g-3p	hsa-miR-1273g-5p
hsa-miR-1275/4665-5p	hsa-miR-1276
hsa-miR-1279	hsa-miR-128-1-5p
hsa-miR-1283	hsa-miR-1284
hsa-miR-1287-3p	hsa-miR-1287-5p
hsa-miR-128b	hsa-miR-129-1-3p
hsa-miR-1290	hsa-miR-1291/6775-3p
hsa-miR-1293/4483	hsa-miR-1294
hsa-miR-1295b-5p/1912	hsa-miR-1296-3p
hsa-miR-1298-5p	hsa-miR-1299
hsa-miR-1302	hsa-miR-1302/4298
hsa-miR-1305	hsa-miR-1306-3p
hsa-miR-130a-3p	hsa-miR-130a-5p

hsa-miR-132-3p/212-3p	hsa-miR-132-5p
hsa-miR-1324	hsa-miR-133a-3p
hsa-miR-134-3p	hsa-miR-134-5p
hsa-miR-135a-3p	hsa-miR-135a-5p
hsa-miR-136-5p	hsa-miR-137
hsa-miR-138-5p	hsa-miR-139-3p
hsa-miR-140-3p.2	hsa-miR-140-5p
hsa-miR-142-3p	hsa-miR-142-3p.1
hsa-miR-143-5p	hsa-miR-144-3p
hsa-miR-146-5p	hsa-miR-1468-3p
hsa-miR-146a-5p	hsa-miR-146b-3p
hsa-miR-147a	hsa-miR-147b
hsa-miR-148a-5p	hsa-miR-148b-3p
hsa-miR-149-3p/4728-5p/6785-5p/6883-5p	hsa-miR-149-5p
hsa-miR-151-3p	hsa-miR-151-5p
hsa-miR-152-5p	hsa-miR-153-3p
hsa-miR-1538	hsa-miR-1538/4745-3p
hsa-miR-155-3p	hsa-miR-155-5p
hsa-miR-15a-5p	hsa-miR-15b-3p
hsa-miR-16-2-3p/195-3p	hsa-miR-16-5p
hsa-miR-17-5p/20-5p/93-5p/106-5p/519-3p	hsa-miR-18-5p
hsa-miR-181a-3p	hsa-miR-181a-5p
hsa-miR-181d-3p	hsa-miR-181d-5p
hsa-miR-1827	hsa-miR-183-3p
hsa-miR-184	hsa-miR-1843
hsa-miR-186-5p	hsa-miR-187-3p
hsa-miR-18a-3p	hsa-miR-18a-5p
hsa-miR-19-5p	hsa-miR-190-5p
hsa-miR-1909-3p/6722-3p	hsa-miR-1909-5p
hsa-miR-191-3p	hsa-miR-191-5p
hsa-miR-1911-3p	hsa-miR-1911-5p
hsa-miR-1914-5p	hsa-miR-1915-3p
hsa-miR-192-5p	hsa-miR-192-5p/215-5p
hsa-miR-193b-3p	hsa-miR-193b-5p
hsa-miR-195-5p	hsa-miR-196-5p
hsa-miR-196b-5p	hsa-miR-197-3p
hsa-miR-1973	hsa-miR-1976
hsa-miR-199a-3p	hsa-miR-199a-5p
hsa-miR-19a-5p	hsa-miR-19b-1-5p
hsa-miR-200a-5p	hsa-miR-200ab-5p
hsa-miR-200c-3p	hsa-miR-200c-5p
hsa-miR-203a-3p	hsa-miR-203a-3p.1
hsa-miR-203b-5p/6718-5p	hsa-miR-204-3p
hsa-miR-205-3p	hsa-miR-205-5p
hsa-miR-206	hsa-miR-208-3p
hsa-miR-20a-3p	hsa-miR-20a-5p
hsa-miR-21-3p/3591-3p	hsa-miR-21-5p
hsa-miR-211-3p	hsa-miR-211-5p
hsa-miR-2114-5p	hsa-miR-2115-3p
hsa-miR-2117	hsa-miR-212-3p
hsa-miR-214-5p	hsa-miR-215-3p
hsa-miR-216b-3p	hsa-miR-216b-5p

hsa-miR-218-5p	hsa-miR-219-5p
hsa-miR-219b-3p	hsa-miR-219b-5p
hsa-miR-221-3p/222-3p	hsa-miR-221-5p
hsa-miR-223-3p	hsa-miR-223-5p
hsa-miR-2276-5p	hsa-miR-2277-3p
hsa-miR-2355-3p	hsa-miR-2355-5p
hsa-miR-23a-5p	hsa-miR-23b-3p
hsa-miR-24-2-5p	hsa-miR-24-3p
hsa-miR-25-3p	hsa-miR-25-3p/32-5p/92-3p/363-3p/367-3p
hsa-miR-2681-3p	hsa-miR-2681-5p
hsa-miR-26a-1-3p	hsa-miR-26a-2-3p
hsa-miR-27-3p	hsa-miR-27a-3p
hsa-miR-28-3p	hsa-miR-28-5p
hsa-miR-2909	hsa-miR-296-3p
hsa-miR-299-3p	hsa-miR-299-5p
hsa-miR-29b-2-5p	hsa-miR-29b-3p
hsa-miR-30-5p	hsa-miR-300
hsa-miR-301b-3p	hsa-miR-301b-5p
hsa-miR-302b-3p	hsa-miR-302b-5p
hsa-miR-302c-5p	hsa-miR-302d-3p
hsa-miR-3064-3p	hsa-miR-3064-5p
hsa-miR-3074-5p	hsa-miR-30a-3p
hsa-miR-30b-5p	hsa-miR-30c-1-3p
hsa-miR-30d-3p	hsa-miR-30d-5p
hsa-miR-31-5p	hsa-miR-3115
hsa-miR-3119	hsa-miR-3120-3p
hsa-miR-3122	hsa-miR-3122/3913-5p
hsa-miR-3125/3916/6859-5p	hsa-miR-3126-3p
hsa-miR-3127-5p	hsa-miR-3128
hsa-miR-3130-3p	hsa-miR-3130-5p
hsa-miR-3134	hsa-miR-3135a
hsa-miR-3136-5p	hsa-miR-3136-5p/4439
hsa-miR-3140-3p	hsa-miR-3140-5p
hsa-miR-3144-5p	hsa-miR-3145-3p
hsa-miR-3148	hsa-miR-3149
hsa-miR-3151-3p	hsa-miR-3151-5p
hsa-miR-3153/6733-5p/6739-5p	hsa-miR-3154
hsa-miR-3156-5p	hsa-miR-3157-3p
hsa-miR-3159	hsa-miR-3160-3p
hsa-miR-3162-5p	hsa-miR-3163
hsa-miR-3167	hsa-miR-3168
hsa-miR-3171	hsa-miR-3173-3p
hsa-miR-3175	hsa-miR-3176
hsa-miR-3178	hsa-miR-3179
hsa-miR-3181	hsa-miR-3182
hsa-miR-3185	hsa-miR-3186-3p
hsa-miR-3189-3p	hsa-miR-3189-5p
hsa-miR-3191-5p	hsa-miR-3192-3p
hsa-miR-3194-5p	hsa-miR-3195
hsa-miR-3198/4309	hsa-miR-3199
hsa-miR-320	hsa-miR-3200-3p
hsa-miR-3202	hsa-miR-320a

hsa-miR-320d
 hsa-miR-323b-3p
 hsa-miR-325
 hsa-miR-328-5p/6885-5p
 hsa-miR-330-3p
 hsa-miR-335-3p
 hsa-miR-338-5p
 hsa-miR-33b-3p
 hsa-miR-340-3p/6827-3p
 hsa-miR-345-3p
 hsa-miR-34b-3p
 hsa-miR-3529-3p
 hsa-miR-3605-5p
 hsa-miR-3609
 hsa-miR-3612
 hsa-miR-3615
 hsa-miR-3618
 hsa-miR-362-5p/500b-5p
 hsa-miR-3622a-3p/3622b-3p
 hsa-miR-363-5p
 hsa-miR-365-3p
 hsa-miR-3653-3p
 hsa-miR-3657
 hsa-miR-365a-5p/365b-5p
 hsa-miR-3663-3p
 hsa-miR-3666
 hsa-miR-367-5p
 hsa-miR-3674
 hsa-miR-3678-3p
 hsa-miR-3680-5p
 hsa-miR-3683
 hsa-miR-3688-3p
 hsa-miR-3689b-3p
 hsa-miR-369-5p
 hsa-miR-3692-5p
 hsa-miR-3714
 hsa-miR-371b-5p/373-5p/616-5p
 hsa-miR-374-5p
 hsa-miR-374c-3p
 hsa-miR-376a-2-5p
 hsa-miR-376c-3p
 hsa-miR-378a-3p
 hsa-miR-378g
 hsa-miR-380-5p
 hsa-miR-382-5p
 hsa-miR-384
 hsa-miR-3911
 hsa-miR-3914
 hsa-miR-3919
 hsa-miR-3923
 hsa-miR-3927-3p
 hsa-miR-3929

hsa-miR-320e
 hsa-miR-323b-5p/410-5p/494-5p
 hsa-miR-325-3p
 hsa-miR-329-3p
 hsa-miR-330-3p.2
 hsa-miR-335-5p
 hsa-miR-339-3p
 hsa-miR-33b-3p/515-3p/519e-3p
 hsa-miR-340-5p
 hsa-miR-345-5p
 hsa-miR-34b-5p
 hsa-miR-3529-5p
 hsa-miR-3606-3p
 hsa-miR-361-3p
 hsa-miR-3613-3p
 hsa-miR-3616-3p
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hsa-miR-99b-3p	hsa-miR-99b-5p

E External-service ledger

Counting conventions and the rejected candidates with evidence are in `EXTERNAL_SERVICES.md`; dual count: 43 counted / 41 most-conservative.

#	Service	Genuine use in this lane	Evidence
1	NCBI GEO (search/FTP/SOFT)	Accession discovery, series/sample SOFT retrieval, matrix download for GSE238208, GSE196156, GSE11839, GSE28242, GSE57560, GSE621, GSE11783	sources/*.txt, sources/*_crosswalk.csv
2	NCBI GEO platform tables (GPL6244, GPL29077 annotation)	Symbol mapping for urine transfer (190/190 M6 symbols)	results/ic_p48_a2_urine_gse28242_result.js
3	Enrichr (Maayan Lab) - MSigDB_Hallmark_2020 library	M0-M6 module gene sets (locked)	prereg/ic_p48_module_genesets.json
4	Enrichr - GO_Biological_Process_2025 library	M3 pain module gene set (GO:0019233, 21 genes, locked)	prereg/ic_p48_module_genesets.json
5	ChatGPT (program consult, redirect mode)	Round 1 redirect after corrected k=2 negative; round 2 residual-framework consult	redirect/round1_*, redirect/round2_*
6	miRTarBase v10 (new host awi.cuhk.edu.cn)	Human validated miRNA-target pairs vs the 200-gene M6 module (87,627 pairs; per-gene CSV export endpoint)	results/ic_p48_a2_urine_gse196156_result.js
7	miRBase (mature.fa)	MIMAT accession -i mature miRNA name map (2,789 hsa entries) for A2.3b	results/ic_p48_a2_urine_gse196156_result.js
8	Ensembl REST	Per-gene lookup for M6 module genes	projects/interstitial_cystitis/results/external
9	UniProtKB REST	Reviewed human accessions per M6 gene	same
10	Europe PMC	"interstitial cystitis" co-mention literature per M6 gene	same
11	MyGene.info	Identifier cross-check per M6 gene	same
12	ChEMBL API	Target lookup per M6 gene	same
13	HGNC REST (gene-names.org)	Symbol status/ID per M6 gene	same
14	GTEx Portal API	Reference gene lookup per M6 gene	same
15	Reactome Content Service	Pathway mapping via UniProt accession per M6 gene	same
16	g:Profiler g:GOST	Enrichment of the committed 200-gene M6 list: top hits mitotic cell cycle (GO:1903047/GO:0000278/GO:0007049) and REAC Cell Cycle - operational validation of the module list	results/external/secondary_annotations_m6 (in this directory)
17	STRING API	Network query among the 5 miRNA-hub M6 genes: 0 physical edges returned (honest null: miRNA hubs are not a physical complex)	same
18	Open Targets Platform GraphQL	Disease search resolved interstitial cystitis = EFO_1000869; hub genes G3BP1/SRSF1/DR1 not among its top-50 associated targets (honest null, recorded)	same
19	EMBL-EBI (MONDO)	OLS Disease ontology anchor MONDO:0018301 "interstitial cystitis"	same
20	QuickGO	GO annotations per hub gene	same
21	IntAct	Interaction records per hub gene	same
22	InterPro	Domain annotations per hub gene	same

#	Service	Genuine use in this lane	Evidence
23	OpenAlex	"hub gene + interstitial cystitis" literature metadata	same
24	Human Protein Atlas	Per-gene tissue/protein context (via Ensembl IDs)	same
25	DGIdb (GraphQL)	Drug-gene interactions per hub: MAPK14 druggable (105 interactions incl. p38 inhibitors); other hubs 0 - honest druggability gradient	results/external/tertiary_annotations_m6.js
26	GWAS Catalog REST v2	Mapped-gene associations per hub (verified 'mapped_gene=' filter with a bogus-gene control; traits recorded per gene)	same
27	NCBI PubMed eutils	Programmatic "interstitial cystitis" Title/Abstract co-mention counts per hub: all 0 - honest null (IC abstracts do not co-mention these hubs)	same
28	Pharos (NIH/NCATS GraphQL)	Target development levels: MAPK14 Tchem/Kinase; G3BP1, DR1, RASAL2 Tbio; SRSF1 Tbio	same
29	cBioPortal API	Hub mutation records in TCGA bladder (blca_tcga_pan_can_atlas_2018): RASAL2 16, SRSF1 6, MAPK14 4, G3BP1 2, DR1 1	same
30	WikiPathways JSON API	Human pathway search per hub (e.g. MAPK14 in 1,131 human pathway text hits)	same
31	PGS Catalog REST	Trait search "cystitis"/"bladder pain syndrome": 0 PGS traits - honest empty	same
32	Monarch Initiative v3	IC disease record (MONDO:0018301) + association predicates + disease phenotypes (bladder/urethra/pain features recorded)	results/external/quaternary_annotations_m6.js
33	NCBI ClinVar (eutils)	Clinical variant record counts per hub (RASAL2 222 ... DR1 20); distinct NCBI database, GEO-vs-PubMed counting precedent	same
34	NCBI dbSNP (eutils)	Human variant counts per hub (RASAL2 145,551 ... SRSF1 9,441); distinct NCBI database	same
35	NCBI Gene (eutils)	Gene records/summaries per hub; distinct NCBI database	same
36	UCSC Genome Browser API	hg38 locus search per hub (position matches recorded)	same
37	Signor	Full-network download (43,570 rows) filtered to human hub edges: MAPK14 239, G3BP1 11, SRSF1 9, DR1 1, RASAL2 0 (per-protein params verified ignored; network filtered locally)	same
38	KEGG REST	Gene records + pathway membership per hub (MAPK14 hsa:1432 in 65 pathways; G3BP1/SRSF1/RASAL2 sparse - honest gradient)	results/external/quinary_annotations_m6.js

#	Service	Genuine use in this lane	Evidence
39	GO Consortium Central API	Term records behind locked modules: GO:0000086 G2/M transition (M6), GO:0019233 sensory perception of pain (M3)	same
40	Rfam	Family records for the top-3 A2.3b regulator miRNAs by validated M6 pair count: mir-16 RF00254, mir-24 RF00178, mir-3613 RF03323	same
41	RNAcentral	RNA record for top regulator hsa-miR-16-5p (URS00004BCD9C)	same
42	Pathway Commons (NCI-PID datasource)	Pathway search per hub restricted to NCI-PID (Reactome excluded to avoid alias): MAPK14 33 hits (top: p38-alpha/beta signaling)	results/external/buffer_annotations_m6.json
43	ClinicalTrials.gov API v2	IC/BPS trial landscape sample + p38 MAPK inhibitor trial search (context for the DGIdb druggability finding)	same

F Per-gene external annotation digest

One line per M6 module gene, from the committed per-gene snapshots; columns are counts returned by each service, not curated judgments.

Gene	UniProt	EuropePMC-IC hits	ChEMBL targets	Reactome pathways
ABL1	P00519	13	17	13
AMD1	P17707	0	3	1
ARID4A	P29374	0	1	2
ATF5	Q9Y2D1	4	0	2
ATRX	P46100	8	1	3
AURKA	O14965	12	10	8
AURKB	Q96GD4	3	8	10
BARD1	Q99728	2	0	21
BCL3	P20749	3	1	
BIRC5	O15392	9	1	10
BRCA2	P51587	32	2	14
BUB1	O43683	3	2	6
BUB3	O43684	1	2	10
CASP8AP2	Q9UKL3	0	0	1
CBX1	P83916	1	1	2
CCNA2	P20248	6	8	21
CCNB2	O95067	5	2	11
CCND1	P24385	28	10	18
CCNF	P41002	0	0	2
CCNT1	O60563	1	5	16
CDC20	Q12834	3	2	20
CDC25A	P30304	7	1	9
CDC25B	P30305	2	2	4
CDC27	P30260	1	3	18
CDC45	O75419	1	1	4
CDC6	Q99741	3	1	8

Gene	UniProt	EuropePMC-IC hits	ChEMBL targets	Reactome pathways
CDC7	O00311	6	4	3
CDK1	P06493	31	17	36
CDK4	P11802	25	24	19
CDKN1B	P46527	10	2	16
CDKN2C	P42773	2	0	4
CDKN3	Q00526	0	6	
CENPA	P49450	1	0	7
CENPE	Q02224	0	8	10
CENPF	P49454	4	0	7
CHAF1A	Q13111	0	0	
CHEK1	O14757	4	2	10
CHMP1A	Q9HD42	0	1	1
CKS1B	P61024	0	3	2
CKS2	P33552	3	1	
CTCF	P49711	9	1	2
CUL1	Q13616	4	3	36
CUL3	Q13618	7	4	11
CUL4A	Q13619	2	10	10
CUL5	Q93034	2	1	6
DBF4	Q9UBU7	0	2	2
DDX39A	O00148	0	1	2
DKC1	O60832	0	1	2
DMD	P11532	27	2	3
DR1	Q01658	5	1	2
DTYMK	P23919	0	1	1
E2F1	Q01094	30	3	17
E2F2	Q14209	3	1	5
E2F3	O00716	5	1	6
E2F4	Q16254	1	1	10
EFNA5	P52803	1	0	3
EGF	P01133	297	144	37
ESPL1	Q14674	1	1	1
EWSR1	Q01844	6	2	
EXO1	Q9UQ84	0	1	17
EZH2	Q15910	39	11	10
FANCC	Q00597	1	1	3
FBXO5	Q9UKT4	0	0	5
FOXN3	O00409	1	0	
G3BP1	Q13283	3	1	1
GINS2	Q9Y248	0	0	1
GSPT1	P15170	3	7	4
H2AX	P16104	26	0	52
H2AZ1	P0C0S5	0	1	2
H2AZ2	Q71UI9	0	0	45
H2BC12	O60814	0	0	58
HIF1A	Q16665	267	6	11
HIRA	P54198	15	6	2
HMGA1	P17096	6	1	7
HMGB3	O15347	0	1	
HMGN2	P05204	1	0	
HMMR	O75330	3	1	3

Gene	UniProt	EuropePMC-IC hits	ChEMBL targets	Reactome pathways
HNRNPD	Q14103	0	1	6
HNRNPU	Q00839	2	1	5
HOXC10	Q9NYD6	2	0	
HSPA8	P11142	4	3	21
HUS1	O60921	0	0	7
ILF3	Q12906	3	1	2
INCENP	Q9NQS7	5	3	7
JPT1	Q9UK76	0	0	
KATNA1	O75449	0	1	
KIF11	P52732	2	4	4
KIF15	Q9NS87	1	1	4
KIF20B	Q96Q89	0	2	2
KIF22	Q14807	1	1	4
KIF23	Q02241	2	1	5
KIF2C	Q99661	2	1	10
KIF4A	O95239	1	1	5
KIF5B	P33176	6	4	7
KMT5A	Q9NQR1	0	2	4
KNL1	Q8NG31	3	0	7
KPNA2	P52292	3	3	9
KPNB1	Q14974	2	3	15
LBR	Q14739	11	1	11
LIG3	P49916	0	1	6
LMNB1	P20700	2	1	10
MAD2L1	Q13257	0	0	11
MAP3K20	Q9NYL2	0	2	
MAPK14	Q16539	6	10	18
MARCKS	P29966	3	1	1
MCM2	P33993	6	3	6
MCM3	P25205	0	2	6
MCM5	P33992	3	1	7
MCM6	Q14566	2	1	6
MEIS1	O00470	5	1	1
MEIS2	O14770	2	0	
MKI67	P46013	14	2	
MNAT1	P51948	1	2	33
MT2A	P02795	2	0	2
MTF2	Q9Y483	0	0	1
MYBL2	P10244	5	1	3
MYC	P01106	135	30	19
NASP	P49321	1	1	
NCL	P19338	24	2	3
NDC80		4	2	0
NEK2	P51955	4	3	10
NOLC1	Q14978	2	1	
NOTCH2	Q04721	13	2	9
NSD2	O96028	2	3	5
NUMA1	Q14980	2	1	2
NUP50	Q9UKX7	1	1	30
NUP98	P52948	4	0	37
NUSAP1	Q9BXS6	3	0	

Gene	UniProt	EuropePMC-IC hits	ChEMBL targets	Reactome pathways
ODC1	P11926	0	4	2
ODF2	Q5BJF6	1	0	7
ORC5	O43913	0	1	7
ORC6	Q9Y5N6	0	0	7
PAFAH1B1	P43034	2	0	14
PBK	Q96KB5	3	1	
PDS5B	Q9NTI5	0	1	4
PLK1	P53350	6	8	25
PLK4	O00444	4	2	7
PML	P29590	24	2	9
POLA2	Q14181	1	2	9
POLE	Q07864	157	5	10
POLQ	O75417	0	1	1
PRC1		6	1	0
PRIM2	P49643	1	1	9
PRMT5	O14744	5	5	3
PRPF4B		0	0	0
PTTG1	O95997	3	0	3
PTTG3P	Q9NZH4	1	0	
PURA	Q00577	8	1	
RACGAP1	Q9H0H5	0	1	12
RAD21	O60216	5	0	7
RAD23B	P54727	1	1	4
RAD54L	Q92698	0	1	
RASAL2	Q9UJF2	0	0	3
RBL1	P28749	3	1	7
RBM14	Q96PK6	0	1	1
RPA2	P15927	3	1	29
RPS6KA5	O75582	1	2	5
SAP30	O75446	0	1	3
SFPQ	P23246	3	1	2
SLC12A2	P55011	3	2	1
SLC38A1	Q9H2H9	3	1	2
SLC7A1	P30825	2	1	1
SLC7A5	Q01650	6	3	3
SMAD3	P84022	48	8	22
SMARCC1	Q92922	2	1	11
SMC1A	Q14683	0	1	7
SMC2	O95347	1	1	2
SMC4	Q9NTJ3	0	1	2
SNRPD1	P62316	0	2	7
SQLE	Q14534	3	3	2
SRSF1	Q07955	9	1	7
SRSF10	O75494	0	0	3
SRSF2	Q01130	2	1	7
SS18	Q15532	5	1	8
STAG1	Q969W9	0	0	1
STIL	Q15468	1	0	
STMN1	P16949	4	1	1
SUV39H1	O43463	4	1	2
SYNCRIP	O60506	2	1	

Gene	UniProt	EuropePMC-IC hits	ChEMBL targets	Reactome pathways
TACC3	Q9Y6A5	2	2	2
TENT4A	Q5XG87	0	1	2
TFDP1	Q14186	0	1	19
TGFB1	P01137	42	7	21
TLE3	Q04726	2	0	5
TMPO	P42166	0	3	
TNPO2	O14787	0	0	
TOP1	P11387	5	8	1
TOP2A	P11388	11	6	2
TPX2	Q9ULW0	2	2	2
TRA2B	P62995	1	1	5
TRAIP	Q9BWF2	0	0	1
TROAP	Q12815	2	0	
TTK		2	3	0
UBE2C	O00762	2	1	19
UBE2S	Q16763	0	2	19
UCK2	Q9BZX2	1	3	1
UPF1	Q92900	2	1	2
WRN	Q14191	3	1	17
XPO1	O14980	5	8	19
YTHDC1	Q96MU7	2	1	1